

Police Analytics App - Local Execution Guide

Step-by-step instructions to run the Django Geo-Incidents application on your local server.

1. Prerequisites

The project is configured with **GeoDjango** and a spatial SQLite engine (**Spatialite**). The necessary external binary dependencies (GDAL, GEOS, mod_spatialite) are bundled in the workspace and mapped dynamically via settings.py. Make sure you use the bundled virtual environment.

2. Activate the Virtual Environment

Open your terminal in the project root directory and run the appropriate activation command for your shell:

PowerShell:

```
> .venv\Scripts\Activate.ps1
```

Command Prompt (cmd):

```
> .venv\Scripts\activate.bat
```

3. Create an Administrator Account (Superuser)

To manage database objects and view incidents in the Django admin dashboard, create a superuser:

```
> python manage.py createsuperuser
```

Follow the interactive prompts to define your admin username, email, and password.

4. Start the Local Server

Run the Django development server using the following command:

```
> python manage.py runserver
```

5. Local Server URL & Entrypoints

Once running, you can access the application at the following local addresses:

Interface	Local URL	Purpose
Dashboard Heatmap	http://127.0.0.1:8000/	Main GIS Map and incident tracker.
Django Admin	http://127.0.0.1:8000/admin/	Database administration interface.
Biometrics Dashboard	http://127.0.0.1:8000/biometrics/	Fingerprint matching and search system.
Analytics Dashboard	http://127.0.0.1:8000/analytics/	Analytical charts & statistics view.

Note on Preloaded Database Data:

The spatial database (db.spatialite) is already preloaded with:

- **37 Nigeria States** (*spatial geometries*)
- **122 Sample Incidents** (*spatial coordinates mapped to States/LGAs*)

No additional setup, migrations, or data import are necessary to get started.